What Contextual Holonomy Detects: the All-versus-Nothing Sector, Two Gap Certificates, and the Section Dependence of Wigner–Friend Holonomy

Manuel Flores Gordillo
Independent Researcher ORCID 0009-0006-8246-0343
manuel.flores@columbia.edu

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Abstract

The slogan "contextual holonomy encodes quantum contextuality" is attractive and, as usually stated, false. We give it a sharp, machine-verified form. On the context stack $[X/\mathrm{PU}(n)]$ of maximal abelian subalgebras the canonical degree-2 class has order exactly n; its cocycle holonomy along closed context loops detects *exactly* the all-versus-nothing (AvN) sector of contextuality for Weyl context families (Theorem 1), a two-directional equivalence stress-tested on 2,100 random families at $d = 2, 4$ with zero mismatches. Both containments are strict: at $d = 3$ the class is nontrivial of order 3 while the achievable state-independent AvN value is 0 (Proposition 1, exhaustive certificate), and KCBS exhibits state-dependent contextuality where the class shadow vanishes (Proposition 2). Finally, the holonomy of the Wigner–friend loop generated by H and S splits by level: its projective (Pancharatnam) holonomy is $+1$, while its determinant-section holonomy is exactly $-i$, a τ-level invariant of the 2-adic tower (Observation 1). Interpretational claims must therefore specify loop and level. We state the residual equivalence problem precisely and give hardware-testable predictions.

1 Setting and the repaired slogan

Let X be the space of maximal abelian $*$-subalgebras (MASAs) of $M_n(\mathbb{C})$, $n \geq 3$. The unitary group acts by conjugation; on $\mathrm{U}(n) \ltimes X$ the $\mathrm{U}(1)$-lifting class is vacuous, and after Morita reduction the canonical class on $\mathrm{PU}(n) \ltimes X$ has order exactly n [1, Thm. 1, Thm. 8] (`thmG_general.py`). Weyl context families and their commutator-carry invariant Q are as in [2]. Throughout, "verified" means: reproduced by the named script in the companion repository [5]; expected one-line outputs are pinned in `verification/INDEX.md`.

The repaired slogan is: *the class detects the AvN sector of contextuality, no more and no less.* The next three results make each clause exact.

2 The equivalence on the AvN sector

Fix $d \geq 2$, $m \geq 1$, and a Weyl context family F whose contexts are commuting triples multiplying to a central element. Write the value-assignment system of F as $A\gamma \equiv s \pmod{d}$, where A

is the context–observable incidence matrix and s collects the matrix-computed context phase exponents under the pinned Weyl convention.

Theorem 1 (Equivalence, AvN sector)**.** *F admits no noncontextual value assignment over $\mathbb{Z}_d$ if and only if some cycle of F carries odd anomaly self-pairing Q.*

The criterion direction is [2, Thm. 4]. The solvability reformulation above is verified two-directionally in `holonomy_vs_solvability.py`: over 1,500 random families at $d = 2$ and 600 at $d = 4$, unsolvability of $A\gamma \equiv s$ (decided by exhaustive dual search) coincided with the odd-Q cycle test in every instance, with the Peres–Mermin square ($d = 2$) and the certified $d = 4$ family of [2] pinned as positives so that both truth values of the equivalence are exercised at both d. The achievable obstruction values themselves are classified as $\{0, d/2\}$ for even d and $\{0\}$ for odd d [2, Thm. 1].

3 Both containments are strict

Proposition 1 (Nontrivial class, zero AvN)**.** *On $\mathbb{Z}_3^2$ the Heisenberg cocycle $\omega(u, v) = \zeta_3^{\langle u,v\rangle}$ and its square are coboundaries for none of the $3^9 = 19{,}683$ cochains; the class has order exactly 3. Simultaneously, every cycle of every sampled qutrit Weyl family has even self-pairing, and the achievable AvN value at $d = 3$ is 0.*

Verified exhaustively in `d3_gap_certificate.py`. Hence a nontrivial canonical class does not imply operational (AvN) contextuality: the weakening adopted for the gerbe-theoretic program [3] is necessary, not cautious.

Proposition 2 (Contextuality, zero class shadow)**.** *The KCBS pentagon in $d = 3$ attains quantum value $\sqrt{5} \approx 2.2361$ against the exhaustive classical bound 2, while the state-independent class shadow at $d = 3$ is 0.*

Verified in `kcbs_converse.py`. Hence state-dependent contextuality can be invisible to the class: the converse direction of the naive slogan also fails.

4 The Wigner–friend loop is level-split

Consider the friend-qubit loop $T \to T_x \to T_y \to T$ generated by $U_1 = \nVdash \otimes H$, $U_2 = \nVdash \otimes S$.

Observation 1 (Section dependence)**.** *The projective (Pancharatnam) holonomy of the loop is $+1$: the two eigenvector strands carry Bargmann phases $e^{\pm i\pi/4}$ (the Bloch-octant $-\Omega/2$ law), whose product is 1, gauge-invariantly. The determinant-section holonomy is exactly $-i$ for every special-unitary closing arrow; $-i = \tau^{-1}$ at $d = 2$, a τ-level invariant of the 2-adic tower, with section-independent square -1.*

Verified in `wf_loop_holonomy.py` (200 gauge randomizations; 500 closers). Any physical or interpretational claim about “the” holonomy of this loop must state the level; the two answers differ.

Hardware predictions. The strand phases $\pm 45°$ and loop product $+1$ are interferometrically measurable with two-qubit Hadamard tests; on the same device the Peres–Mermin loop yields context signs $(+,+,+,+,+,-)$, i.e. ray-level holonomy -1. One machine, two loops, opposite holonomies, both sharp.

Measured (`ibm_fez`, 2026-07-09, job `d986q62f47jc73a7hm2g`; registry record `hardware/results/ibm_fez_hol` strand phases $+45.88°$ and $-48.81°$ against targets $\pm 45°$ (per-strand statistical error $\approx 1.1°$; the $\sim 3.5\sigma$ offset on the X-prepared strand is a plausible preparation systematic), loop phase $-2.93° \pm 1.55°$ against target $0°$. The ray-level holonomy of the Wigner–friend loop is thus measured to be $+1$; the value $-i$ (loop $-90°$) is excluded at this level by $\approx 56\sigma$, in accordance with Observation 1: $-i$ is a section-level, not interferometric, invariant. On the same device the Peres–Mermin loop gave context products $(+0.750, +0.835, +0.722, +0.780, +0.824, -0.700)$, sign pattern 6/6 as predicted, and $S = 4.6125 \pm 0.0173$: a 35.4σ violation of the exact deterministic noncontextual bound 4 (the run-time script carried the loose $n_c - 1 = 5$ bound, since replaced in `qkernel` by the exhaustive tight bound), consistent with the archived `ibm_fez` value 4.558 ± 0.018.

5 The residual problem

Open Problem 1. *Characterize, in invariants of the context stack* $[X/\mathrm{PU}(n)]$ *refined beyond the canonical* H^2 *class, the property "some state has positive contextual fraction on* F*". Equivalently: identify the cohomological home of KCBS-type (state-dependent, class-invisible) contextuality, and prove or refute that a sheaf-theoretic refinement restores a two-directional equivalence.*

Propositions 1–2 show no formulation at the level of the canonical class alone can succeed; Theorem 1 shows the AvN sector is already optimally captured there.

Lemma 1 (No fully covariant qubit-pair nets)**.** *There is no two-qubit quasi-probability frame, covariant under all Pauli displacements, whose every Lagrangian line-sum is a stabilizer projector: the* 15 *Lagrangian sign conditions have rank* 10 *over* $\mathbb{F}_2$ *and are inconsistent, with minimal certificate a Peres–Mermin-type magic configuration (nine observables, each in exactly two of six contexts,* ε*-product* -1*). The order-*2 *class thus obstructs the global frame itself (*`ghw_net_necessity.py`*).*

Genuine Gibbons–Hoffman–Wootters nets evade Lemma 1 by retreating to GF(4) translation covariance; sweeping all 1024 of them (`gf4_net_necessity.py`) eliminates global frames in both remaining directions: a state with contextual fraction 0.3462 on the trivial-class family admits a nonnegative representation in some net (necessity fails), while $T \otimes T$ is negative in every net (best case $-1/4$) with contextual fraction 0 (sufficiency fails); all 60 stabilizer states are nonnegative in some net (construction sanity). Together with the 2^{15}-frame sweep (`net_robust_negativity.py`), every global-frame formulation of the even-d classifier is refuted by explicit certificate; the scenario-paired formulation below is not merely the surviving candidate but the forced one.

The constructive half now exists (`paired_frame_construction.py`). For a closed-triple Weyl family F at $d = 2$, the quantum model $e(\rho)$ is a bijection of the characteristic vector $c(\rho) = (\operatorname{tr} \rho T_v)_{v \in O(F)}$, and every noncontextual model is supported on the solution set $L(F)$ of the family's sign system (parity-violating outcomes have $e \equiv 0$). Hence:

Theorem 2 (Codeword-polytope classification; machine-validated)**.** *For closed-triple Weyl families at $d = 2$, $\mathrm{CF}(F, \rho) = 0 \iff c(\rho) \in \operatorname{conv} L(F)$, the ε-shifted codeword polytope of the family's $\mathbb{F}_2$ kernel; and $L(F) = \varnothing$ exactly on the AvN sector, so one statement covers both sectors. On the two trivial-class Peres–Mermin subfamilies the polytope has exactly* 24 *facets, all triangles $\pm c_u \pm c_v \pm c_w \leq 1$; the classification holds* 40/40 *on both, while the canonical visible frame alone (*15/40, 20/40*) and the induced CHSH squares (*31/40, 29/40*) are each insufficient — negative results retained as certificates.*

The pairing of Conjecture 1 is thus, in the even-d closed-triple case, membership in a polytope cut out by the same $\mathbb{F}_2$ kernel that classifies the AvN sector; what remains open is general d, mixed families, and the written proof of the facet structure.

The ghost facet theorem and a closed formula

The facet structure now has a proof and yields an exact formula (`ghost_facet_theorem.py`). Let F be Peres–Mermin minus one context C^* with sign ε^*. *Bijection lemma:* for a closed triple with $T_u T_v = \varepsilon_C T_w$, the joint distribution is supported on $s_u s_v s_w = \varepsilon_C$ where $e_C(s) = \frac{1}{4}(1 + \sum_i s_i c_i)$; hence $e(\rho) \leftrightarrow c(\rho)$ affinely. *Support lemma:* noncontextual models put no mass on parity-violating outcomes, so they are mixtures over $L(F)$, and a mixture's model depends only on its mean, giving Theorem 2. *Two-tetrahedra lemma:* every $\lambda \in L$ has ghost parity $\prod_{v \in C^*} \lambda_v = -\varepsilon^*$ (the product of the five remaining signs, $= -\varepsilon^*$ because the six Peres–Mermin signs multiply to -1 — the class), while every quantum state's ghost coordinates lie in the opposite tetrahedron of parity ε^* (since $T_1 T_2 T_3 = \varepsilon^* \mathbb{1}$ on C^*). Consequently, for σ with $\prod \sigma_i = \varepsilon^*$, every $\lambda \in L$ disagrees with σ in an odd number of positions, so $\sum_i \sigma_i \lambda_{v_i} \leq 1$: these four inequalities are valid, tight on 12 of the 16 vertices, and are the only quantum-operative facets; the 20 in-context facets are the probability positivity conditions $e_C(s) \geq 0$, never violated by quantum states, and the opposite-parity ghost directions are unconstrained (L reaches 3). The classification therefore collapses to a formula:

$$\mathrm{CF}(F, \rho) \; = \; \max\Big(0, \; \tfrac{S^*(\rho) - 1}{2}\Big), \qquad S^*(\rho) = \max_{\prod \sigma = \varepsilon^*} \sum_{v \in C^*} \sigma_v \operatorname{tr}(\rho T_v),$$

verified to 10^{-16} across 60 states on each of three deletions. The upper bound $\mathrm{CF} \leq (S^* - 1)/2$ follows analytically from the validity above; equality now follows from the *Completion Lemma*: for each operative pattern σ the point y_σ with ghost coordinates σ and all others zero is a valid parity-supported no-signalling model (every remaining context contains at most one ghost observable, so its probabilities are $(1 \pm 1)/4$ or $1/4$), attaining $\sigma \cdot c = 3$; Abramsky–Barbosa–Mansfield duality then gives $\mathrm{CF} = \max_\sigma (S_\sigma - 1)/(3 - 1)$ exactly. The $d = 2$ theorem is thus proved end to end. Corollary: joint eigenstates of C^* with an operative pattern attain $S^* = 3$ and hence $\mathrm{CF} = 1$ — for the deleted odd column these are the Bell states, so *maximal contextuality survives the deletion of the very context that carried the class*: the class does not disappear, it flips the ghost tetrahedron.

The ghost at $d = 4$: the tower appears

The mechanism lifts to the first nontrivial tower level (`d4_ghost_tower.py`), on the certified $d = 4$ magic square (six contexts, nine observables of mixed orders 2 and 4, each observable in exactly two contexts, phase exponents summing to $d/2$ — the class value). Deleting any context

C^* leaves a solvable system with $|L| = 64$ spectral assignments, and the ghost value $\sum_{v\in C^*} \lambda_v$ is constant on L with

$$\gamma - s^* \;\equiv\; d/2 \pmod{d} :$$

the classical ghost torus is rotated from the quantum one by exactly the top 2-adic layer $\omega^{d/2} = -1$, for both a pure order-2 ghost and a mixed order-2/4 ghost — and the $d = 2$ opposite-tetrahedra lemma is the same law, since there too the offset was $d/2$. Every joint eigenstate class of the deleted context attains CF = 1.0000 exactly (4 and 8 classes respectively), and Haar-random states populate the sector (sampled CF up to 0.21). Two negative results are pinned honestly: noncontextual assignments must be restricted to spectral values (non-spectral assignments hit zero-probability outcomes; the unrestricted relaxation under-computes CF), and the $d = 4$ law is *not* a function of the Hermitian order-2 shadows of the ghost alone — the Hermitianized $d = 2$ formula predicts zero precisely where random states reach CF = 0.138. The exact $d = 4$ classification is polytope membership in the full moment space of Theorem 2. Two further results pin its structure (`d4_moment_facets.py`).

Ghost-law lemma (now proved). Since every observable lies in exactly two contexts, summing the remaining constraints over any $\lambda \in L$ gives $\gamma = \sum_{\text{rem}} s - 2T \pmod{d}$, with $T = \sum_{v\notin C^*} \lambda_v \bmod 2$; using $\sum_{\text{all}} s = d/2$ (the class) this reads $\gamma - s^* = d/2 - 2(s^* + T)$. T is constant on L and $s^* + T$ is even for every one of the six deletions (machine-verified), so $\gamma - s^* = d/2$ always: the tower ghost law is an identity plus a verified parity sub-lemma.

τ-necessity (twelve explicit certificates). For each sampled state with CF > 0, the *optimal* separating functional restricted to the even-character subspace — all order-2 shadow data, i.e. everything visible below the top 2-adic layer — achieves gap ≤ 0: the even projection of the quantum moment vector lies *inside* the even projection of the classical polytope. Hence the $d = 4$ state sector is invisible at the order-2 layer, and the τ-level moments are necessary for every operative facet. The 2-adic tower is not bookkeeping; it is load-bearing for the state sector, exactly as the even-d half of Conjecture 1 asserts. The odd-sector facet structure is now partially decoded (`d4_odd_sector_facets.py`). The polytope has affine dimension 33; sparse certificate mining shows the operative geometry is two-tiered. *Tier one* consists of triangle inequalities on *closed τ-twisted order-2 triples*: every 3-sparse certificate coordinate (21/21) is an order-2 operator reached through odd character indices of its host context — the shadow wearing the top-layer cocycle phase — and the recurring triples are the deleted context C^* itself (the τ-twisted ghost) and a virtual closed triple $\{(0, 2, 2, 0), (2, 0, 0, 2), (2, 2, 2, 2)\}$. This tier classifies 32/40 sampled states. *Tier two*, required by the remaining contextual states, involves genuinely order-4 operator moments — the second level of the tower — and is now pinned exactly (`d4_tier2_catalogue.py`): the extracted supporting inequalities have tight-vertex rank exactly 32 (true facets of the 33-dimensional hull), their normals snap with *zero* error onto the $\frac{1}{2\sqrt{2}}\mathbb{Z}$ grid with a single coefficient magnitude $3/(2\sqrt{2})$, and their bounds are the integers 6 and 7 — equivalently, in unit-coefficient normalization, $4\sqrt{2}$ and $14\sqrt{2}/3$: *Tsirelson's $\sqrt{2}$ sits in the tier-two bounds*, with $\tau = e^{i\pi/4}$-quantized phases in the normals. Together, tier one and the exact tier-two facets classify 40/40 states of the construction sample; a subsequent fresh-sample audit shows this is *not yet the complete family* — new weak-band contextual states require further tier-two orbit members. Closing the extracted seeds under the two proven vertex symmetries (kernel translations and global conjugation, $\sim$1152 facets) recovers most but not all fresh states, and per-context outcome relabelings are provably *not* symmetries (each observable lives in two contexts). The deleted family's automorphism group is now derived: the family is $K_{3,3}$ (contexts as nodes, observables as edges), and the automorphisms fixing the deleted context

form the twelve graph automorphisms dressed by a 64-torsor of spectral shifts, times conjugation — order 1536, each element provably a bijection of the 64 vertices, acting on moment space by exact monomial rotations. The census is now *complete and exact*. The polytope — 64 integer $\{-1, 0, 1\}$ points in faithful 33-dimensional integer coordinates — has exactly **23,256** facets in **61** symmetry classes, derived twice by independent methods that agree class-for-class: an adjacency decomposition (breadth-first closure of the facet–ridge graph modulo the derived group, every pivot certified by exact integer nullspace recovery through rank-32 tight sets and exact validity on all vertices), and a direct exact enumeration of the full polytope by NORMALIZ. Tight-vertex counts run 36–60; orbit sizes run 4 to 768, so the census contains both highly symmetric facets (stabilizers of order 192) and free ones. Since quantum moment vectors lie in the affine hull (residual $\sim 10^{-14}$), the catalogue is the classification theorem in operational form: $\mathrm{CF}(\rho) > 0$ if and only if $\mu(\rho)$ violates one of the 23,256 inequalities; on fresh sampled states the facet classifier agrees with the LP 40/40, as it must. Two method notes are pinned in the suite: `lrs` wedges reproducibly on several of these tight-set polytopes (stalling at identical partial output under different budgets) while `cdd` and NORMALIZ succeed; and an intermediate in-walk count of "121 classes" was a bookkeeping duplication (two key encodings double-storing each orbit) exposed only by the independent cross-check — the cross-check is not optional. The tower appears three times over: as the ghost rotation $d/2$, as the stratification of operative facets by operator order, and as the $\mathbb{Z}[1/\sqrt{2}]$ arithmetic of the tier-two facet data. Open: the no-signalling-normalized closed formula, mixed families, and the arithmetic profile of all 61 classes.

Conjecture 1 (State-sector pairing)**.** *For odd prime d, the state-dependent sector is classified by the pairing of the state against the order-d class through the Weyl transform: a state yields positive contextual fraction on stabilizer measurement scenarios if and only if its parity-frame quasi-distribution is negative — the Howard–Wallman–Veitch–Emerson contextuality–negativity correspondence [4], recast as a stack pairing. For even d, where no Gross frame exists, the corresponding role is played by the top layer of the 2-adic tower of [2].*

Computed anchors (`state_sector_probe.py`): $\mathrm{CF(PM)} = 1.0000$ for every state (AvN); $\mathrm{CF(KCBS)} = 0.4721 = 2\sqrt{5} - 4$ at the optimal state with class shadow 0; at $d = 3$ the parity-frame quasi-distribution is nonnegative on all twelve stabilizer pure states (exhaustive) and attains $-1/3$ on the strange state. Notably, the τ-convention frame of [2] exhibits $-1/18$ pseudo-negativity on stabilizer states at $d = 3$: τ is the even-d object, and the odd/even divide of [1] resurfaces at the level of quasi-probability frames.

A Claims-to-verification table

Claim	Script	Expected one-liner
Thm. 1	`holonomy_vs_solvability.py`	0 mismatches; pinned PM/cert4 agree
Prop. 1	`d3_gap_certificate.py`	order 3; AvN $\{0\}$; gap certified
Prop. 2	`kcbs_converse.py`	$2.23607 > 2$; converse gap certified
Obs. 1	`wf_loop_holonomy.py`	strands $\pm\pi/4$; det-holonomy $-i$
PM anchor	`pm_gauge_invariance.py`	product -1 over all 512 gauges

References

[1] M. Flores Gordillo, *The Canonical Class of the MASA Context Stack and the Peres–Mermin Obstruction*, working draft, 2026.

[2] M. Flores Gordillo, *The Obstruction Spectrum of Weyl Context Families and the 2-adic Tower*, working draft, 2026.

[3] M. Flores Gordillo, *Contextual Holonomy, Lifting Bundle Gerbes, and the Geometric Structure of Quantum Contextuality*, in preparation.

[4] M. Howard, J. Wallman, V. Veitch, J. Emerson, *Contextuality supplies the magic for quantum computation*, Nature 510, 351 (2014).

[5] `github.com/manuflog/contextuality-obstructions`, verification suite, 2026.